Finite-q Corrections to the Carnot Capacity Identification

LC-O3 of the Lorentz–Capacity Sub-Programme

Jérôme Beau

Independent Researcher

jerome.beau@cosmochrony.org

2026

Abstract

We close LC-O3, the finite-q correction problem near the saturation boundary of the Lorentz-capacity construction. The quadratic Born–Infeld capacity identity

$$B_n^2 + (F_n^\tau)^2 = 1$$

is not corrected: it is exact by the definition $F_n^\tau := \sqrt{1 - B_n^2}$, provided B_n is the exact finite-q torus load. The correction concerns the pre-saturation Carnot–Carathéodory parabolic identification

$$B_n^{\mathrm{parab}} = \frac{2n^2 + 2n + 1}{q^2},$$

which is exact only before cyclic folding, for $n \leq R := (q-1)/2$. Beyond the folding boundary, writing $n = R + m$, the exact torus count gives

$$B_n^{\mathrm{torus}} = \frac{q(2m+1) + 2R^2 - 2m^2}{q^2},$$

and the deviation from the parabolic Carnot formula is

$$\Delta_q(n) := B_n^{\mathrm{parab}} - B_n^{\mathrm{torus}} = \frac{4m^2}{q^2}.$$

In the macroscopic folded regime $n/q \to \alpha > 1/2$, the correction $\Delta_q(n) \to 4(\alpha - 1/2)^2$ is finite and the parabolic identification is not asymptotically valid. In the pre-saturation and sublinear boundary

regimes ($m/q \to 0$), the correction vanishes and the Lorentz-capacity law is recovered exactly. LC-O3 therefore identifies the finite-q domain of validity of the Carnot approximation without modifying the quadratic capacity law.

Keywords. finite-q corrections; Carnot–Carathéodory identification; saturation boundary; quadratic capacity law; torus folding; Born–Infeld admissibility; Cosmochrony.

Contents

1 Introduction **4**

2 Pre-saturation exactness **4**

3 Exact torus formula after cyclic folding **5**

4 The finite-q correction **6**

5 Domain of validity of the Carnot identification **6**

6 Asymptotic recovery of the Lorentz-capacity law **7**

7 Conclusion **8**

1 Introduction

The Lorentz-capacity sub-programme derives the temporal residual $F_n^\tau = \sqrt{1 - B_n^2}$ from the bounded Born–Infeld capacity budget of the admissibility cascade on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Papers [1] and [2] prove that the quadratic identity

$$B_n^2 + (F_n^\tau)^2 = 1$$

holds exactly for all $n < q/2$, using the Weil filling law $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ and the temporal residual map $F_n^\tau = P_\tau^{Q5b} \Sigma_n^{\mathrm{tot}} = \sqrt{1 - B_n^2}$. This closes time dilation and the Lorentz factor identification.

The open problem LC-O3 (Section 7.2 of [2]) is the status of the quadratic law and the Carnot capacity identification near the finite-q saturation boundary $n \simeq q/2$.

The resolution is a precise distinction between two statements:

1. The quadratic identity $B_n^2 + (F_n^\tau)^2 = 1$ is exact for every n, once B_n is the exact finite-q torus load. It requires no correction.

2. The parabolic Carnot formula $B_n^{\mathrm{parab}} = (2n^2 + 2n + 1)/q^2$ is exact only for $n \le R := (q-1)/2$. Beyond the folding boundary, it deviates from the exact torus count by $\Delta_q(n) = 4m^2/q^2$ where $m = n - R$.

Thus LC-O3 is a correction to the Carnot approximation, not to the capacity law.

Section 2 recalls the pre-saturation exactness. Section 3 derives the exact torus formula for $n \ge R$. Section 4 proves the correction theorem. Section 5 analyses the three asymptotic regimes. Section 6 states the asymptotic recovery corollary.

2 Pre-saturation exactness

Let q be an odd prime and set $R := (q-1)/2$. For $a \in \mathbb{Z}/q\mathbb{Z}$, define the cyclic distance $r(a) := \min(a \bmod q,\ q - a \bmod q) \in \{0, 1, \ldots, R\}$. The BFS spatial coverage at depth n is

$$B_n^{\mathrm{torus}} := \frac{1}{q^2} \left| \left\{ (a,b) \in (\mathbb{Z}/q\mathbb{Z})^2 : r(a) + r(b) \le n \right\} \right|.$$

Lemma 2.1 (Pre-saturation parabolic formula)**.** *For $0 \le n \le R$, the torus count coincides with the unfolded Carnot diamond:*

$$B_n^{\mathrm{torus}} = B_n^{\mathrm{parab}} := \frac{2n^2 + 2n + 1}{q^2}.$$

Proof. For $n \le R$, the diamond $\{(a,b) : r(a) + r(b) \le n\}$ does not wrap around the cyclic identifications in $(\mathbb{Z}/q\mathbb{Z})^2$. Each element of the diamond has a unique pre-image in $\{0, \dots, q-1\}^2$, so the torus count equals the unfolded count $|D_n| = 2n^2 + 2n + 1$, which is the cardinality of the unfolded ℓ^1 diamond of radius n in $\mathbb{Z}^2$. □ □

By the Weil factorization [1], $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ exactly. The temporal residual map of [2] then gives $F_n^\tau = \sqrt{\Sigma_n^{\mathrm{tot}}(2 - \Sigma_n^{\mathrm{tot}})} = \sqrt{1 - B_n^2}$. The identity $B_n^2 + (F_n^\tau)^2 = 1$ is exact for every n, not only in the pre-saturation regime. The pre-saturation lemma is needed for a different reason: to identify the regime where $B_n = B_n^{\mathrm{parab}}$, i.e. where the Carnot formula may be used.

3 Exact torus formula after cyclic folding

For $n > R$, the ℓ^1 ball on $(\mathbb{Z}/q\mathbb{Z})^2$ wraps around the torus.

Lemma 3.1 (Exact torus count after folding)**.** *For $n = R + m$ with $0 \le m \le R$, the exact torus capacity load is*

$$B_n^{\mathrm{torus}} = \frac{q(2m+1) + 2R^2 - 2m^2}{q^2}.$$

Proof. Write the count as

$$\big|\{(a,b) : r(a) + r(b) \le n\}\big| = \sum_{k=0}^{R} \#\{a : r(a) = k\} \cdot \#\{b : r(b) \le n - k\}.$$

The single-coordinate counts are $\#\{a : r(a) = 0\} = 1$ and $\#\{a : r(a) = k\} = 2$ for $1 \le k \le R$; and $\#\{b : r(b) \le j\} = \min(2j+1,\, q)$.

For $n = R + m$, the bound $n - k \ge R$ holds iff $k \le m$, so $\min(2(n-k)+1, q) = q$ for $k \le m$ and $= 2(n-k)+1$ for $k > m$. Splitting the sum at $k = m$:

$$\big|\{r(a) + r(b) \le n\}\big| = q + 2mq + 2\sum_{k=m+1}^{R} (2(R+m-k)+1).$$

Substituting $j = R - k$ (ranging from $R - m - 1$ down to 0):

$$2\sum_{j=0}^{R-m-1} (2(m+j)+1) = 2\big[2 \cdot \tfrac{(R-m-1)(R-m)}{2} + (2m+1)(R-m)\big] = 2(R^2 - m^2).$$

Therefore the total is $q(2m+1) + 2R^2 - 2m^2$, and dividing by q^2 gives the stated formula. □

Remark 3.2 (Endpoint verification). At $m = 0$ ($n = R$): $B_R^{\text{torus}} = (q + 2R^2)/q^2 = (2R^2 + 2R + 1)/q^2 = B_R^{\text{parab}}$, confirming continuity at the folding boundary. At $m = R$ ($n = q - 1$): $B_{q-1}^{\text{torus}} = (q(2R + 1) + 2R^2 - 2R^2)/q^2 = q^2/q^2 = 1$, confirming that the whole horizontal torus is filled at the final depth.

4 The finite-q correction

Theorem 4.1 (Finite-q correction to the Carnot capacity profile)**.** *For $n = R + m$ with $0 \le m \le R$, the deviation of the parabolic Carnot formula from the exact torus count is*

$$\Delta_q(n) := B_n^{\text{parab}} - B_n^{\text{torus}} = \frac{4m^2}{q^2}.$$

Proof. With $n = R + m$ and $q = 2R + 1$, the parabolic numerator expands as

$$2(R + m)^2 + 2(R + m) + 1 = 2R^2 + 4Rm + 2m^2 + 2R + 2m + 1.$$

The torus numerator, using $q = 2R + 1$, expands as

$$q(2m + 1) + 2R^2 - 2m^2 = 2R^2 + 4Rm + 2R + 2m + 1 - 2m^2.$$

Their difference is $4m^2$, and dividing by q^2 gives $\Delta_q(n) = 4m^2/q^2$. □

Remark 4.2 (Parabolic formula becomes unphysical). Since $B_n^{\text{torus}} \le 1$ always, the correction formula implies that $B_n^{\text{parab}} > 1$ once $4m^2/q^2 > 1 - B_R^{\text{parab}}$, i.e. for sufficiently large m. Table 1 illustrates this for $q \in \{11, 13\}$: the parabolic formula exceeds 1 as soon as $n \ge R+3$ for $q = 11$ and $n \ge R+3$ for $q = 13$. This is the clearest manifestation that B_n^{parab} is not a valid load in the folded regime.

5 Domain of validity of the Carnot identification

The correction $\Delta_q(n) = 4m^2/q^2$ determines three asymptotic regimes.

Pre-saturation regime. For $n \le R$, $m = 0$ and $\Delta_q(n) = 0$. The parabolic Carnot formula is exact.

Table 1: Parabolic vs. exact torus capacity load and correction $\Delta_q \cdot q^2 = 4m^2$, for $q = 11$ (left, $R = 5$) and $q = 13$ (right, $R = 6$). A dagger (†) marks entries where $B_n^{\text{parab}} > 1$.

$q = 11$				$q = 13$			
n	m	B_n^{parab}	B_n^{torus}	n	m	B_n^{parab}	B_n^{torus}
5	0	0.504	0.504	6	0	0.503	0.503
6	1	0.702	0.669	7	1	0.669	0.645
7	2	0.934	0.802	8	2	0.858	0.763
8	3	$1.198^{\dagger}$	0.901	9	3	$1.071^{\dagger}$	0.858
9	4	$1.496^{\dagger}$	0.967	10	4	$1.308^{\dagger}$	0.929
10	5	$1.826^{\dagger}$	1.000	11	5	$1.568^{\dagger}$	0.976
				12	6	$1.852^{\dagger}$	1.000

Sublinear boundary layer. For $n = R + m$ with $m/q \to 0$,

$$\Delta_q(n) = \frac{4m^2}{q^2} \to 0.$$

The parabolic formula remains asymptotically valid in a sublinear neighbourhood of the folding boundary.

Macroscopic folded regime. For $n/q \to \alpha > 1/2$, one has $m/q \to \alpha - 1/2$, and therefore

$$\Delta_q(n) \to 4\big(\alpha - \tfrac{1}{2}\big)^2 > 0.$$

The correction does not vanish. The parabolic Carnot identification is not asymptotically valid in this regime.

6 Asymptotic recovery of the Lorentz-capacity law

The exact Lorentz-capacity identity

$$B_n^2 + (F_n^\tau)^2 = 1, \qquad F_n^\tau = \sqrt{1 - B_n^2},$$

holds for every n once $B_n = B_n^{\text{torus}}$. In the pre-saturation regime, $B_n = B_n^{\text{parab}}$ and the Carnot approximation may be used directly. In the folded regime, the correct entry is

$$B_n^{\text{torus}} = B_n^{\text{parab}} - \frac{4m^2}{q^2},$$

and the temporal residual remains exactly $F_n^\tau = \sqrt{1 - (B_n^{\text{torus}})^2}$.

Corollary 6.1 (Asymptotic Carnot recovery)**.** *The parabolic Carnot identification $B_n = B_n^{\text{parab}}$ is asymptotically recovered whenever $m/q \to 0$, i.e. whenever $n \le R$ or $n = R + o(q)$. It is not asymptotically recovered for $n/q \to \alpha > 1/2$. The Lorentz-capacity identity $B_n^2 + (F_n^\tau)^2 = 1$ holds exactly in all regimes with the exact torus load.*

Proof. The first two claims follow from $\Delta_q(n) = 4m^2/q^2 \to 0$ iff $m/q \to 0$. The third claim is the tautological identity $(B_n^{\text{torus}})^2 + (1 - (B_n^{\text{torus}})^2) = 1$. □

7 Conclusion

LC-O3 is closed by distinguishing two statements that were conflated in the open problem.

The first statement is exact and receives no correction: the quadratic Lorentz-capacity identity

$$B_n^2 + (F_n^\tau)^2 = 1$$

holds for every n, with B_n the exact finite-q torus load.

The second statement is approximate and has a precise domain of validity: the parabolic Carnot formula $B_n^{\text{parab}} = (2n^2 + 2n + 1)/q^2$ is exact only for $n \le (q-1)/2$, and deviates from the exact torus count by $\Delta_q(n) = 4m^2/q^2$ for $n = (q-1)/2 + m$. The correction vanishes in the pre-saturation and sublinear boundary regimes, but converges to $4(\alpha - \frac{1}{2})^2 > 0$ in the macroscopic folded regime $n/q \to \alpha > 1/2$.

With LC-O1, LC-O2-O1, and LC-O3 now closed, the Lorentz-capacity sub-programme is complete. Time dilation, the light cone, Lorentz transformations and length contraction, the gravitational capacity–metric bridge, and the finite-q saturation domain of validity are all established.

References

[1] J. Beau. Spectral Mobility and the Lorentz Factor from Projective Capacity Saturation, 2026. Cosmochrony technical note N, doi:10.5281/zenodo.20299173.

[2] J. Beau. Temporal Residual Map from the Completed Principal Symbol: Construction, Uniqueness, and the Lorentz Identification, 2026. Lorentz Capacity sub-programme: temporal residual map construction, doi:10.5281/zenodo.20316527.